Network Configuration Management System

Software Requirements Specification

IEEE 830-1998 outline

DRAFT — human review required

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Contents

1 Introduction 4
1.1 Purpose . . . 4
1.2 Scope . . . 4
1.3 Definitions, acronyms and abbreviations . . . 4
1.4 References . . . 4
1.5 Overview . . . 4
2 Overall description 4
2.1 Product perspective . . . 4
2.2 Product functions . . . 5
2.3 User characteristics and stakeholders . . . 5
2.4 Constraints . . . 5
2.5 Assumptions and dependencies . . . 5
3 Specific requirements 6
3.1 External interface requirements . . . 6
3.1.1 User interfaces . . . 6
3.1.2 Hardware interfaces . . . 6
3.1.3 Software interfaces . . . 6
3.1.4 Communications interfaces . . . 6
3.2 Functional requirements . . . 6
3.3 Performance requirements . . . 7
3.4 Logical database requirements . . . 7
3.5 Design constraints . . . 7
3.6 Software system attributes . . . 7
3.6.1 Reliability . . . 7
3.6.2 Availability . . . 8
3.6.3 Security . . . 8
3.6.4 Maintainability . . . 8
3.6.5 Portability . . . 8
3.7 Other requirements . . . 8
A Derived threat hypotheses 8
A.1 THR-001: User identities and authenticated access . . . 8
A.2 THR-002: Role and permission boundaries . . . 9
A.3 THR-003: Integration trust paths . . . 9
A.4 THR-004: Security and administrative audit evidence . . . 9
A.5 THR-005: Monitoring and security alerts . . . 9
B Risk assessment 10
B.1 RISK-001 . . . 10
B.2 RISK-002 . . . 10
B.3 RISK-003 . . . 10
B.4 RISK-004 . . . 11
B.5 RISK-005 . . . 11
C SDLC recommendation and delivery workflow 11
C.1 Ranking and source evidence . . . 12
C.1.1 Incremental suitability reasons . . . 12
C.1.2 Hybrid suitability reasons . . . 13
C.1.3 DevSecOps suitability reasons . . . 13
C.1.4 Spiral suitability reasons . . . 13
C.1.5 Agile suitability reasons . . . 13
C.1.6 V-Model suitability reasons . . . 13

C.1.7 Waterfall suitability reasons 13
C.2 Methodology phases 13
C.2.1 Requirements and verification planning 13
C.2.2 Incremental delivery cycle 14
C.2.3 Release and handoff 16

1 Introduction

1.1 Purpose

This specification records the project purpose, stakeholder constraints and unresolved decisions stated in the source evidence below.

This SRS describes the software's required behavior for users, administrators and the Project Analyst reviewing the specification.

1.2 Scope

Design a secure network configuration management system for a university that manages configuration changes across routers, switches, wireless controllers, and other network infrastructure. Authorized network administrators should be able to review current configurations, submit configuration changes, track change history, and restore previously approved configurations when necessary. The system should detect unauthorized or unexpected configuration changes, generate security alerts, maintain audit logs of configuration activity and administrator actions, and restrict administrative functions based on approved roles. The platform should provide a centralized dashboard, integrate with existing university identity services, support multiple campus locations, and remain secure, reliable, scalable, and manageable.

1.3 Definitions, acronyms and abbreviations

Term	Meaning
Audit record	A stored record of an event. Required event scope and retention are determined by the linked source, not this definition.
Log	A collection of event records. This term does not itself require additional logging fields, events or access permissions.
Access restoration	Returning access that was previously restricted, within the source-defined permission boundary.

1.4 References

Reference	Source
Format	IEEE Std 830-1998, clauses 5 and Annex A (document organization).

1.5 Overview

Section 2 summarizes the product, stakeholders and open decisions. Section 3 contains the uniquely identified requirements. Detailed evidence, acceptance procedures are available in the Normal report. Appendices A-C contain derived threat hypotheses, risk assessment and the proposed SDLC workflow.

2 Overall description

2.1 Product perspective

Design a secure network configuration management system for a university that manages configuration changes across routers, switches, wireless controllers, and other network infrastructure.

2.2 Product functions

Functional topics are summarized below; section 3 retains the full requirements.

- manage configuration changes across routers, switches, wireless controllers, and other network infrastructure - FR-001
- allow Authorized network administrators to review current configurations - FR-002
- allow Authorized network administrators to submit configuration changes - FR-003
- allow Authorized network administrators to track change history - FR-004
- allow Authorized network administrators to restore previously approved configurations when necessary - FR-005
- detect unauthorized or unexpected configuration changes - FR-006
- generate security alerts - FR-007
- maintain audit logs of configuration activity and administrator actions - FR-008
- restrict administrative functions based on approved roles - FR-009
- provide a centralized dashboard - FR-010
- integrate with existing university identity services - FR-011
- support multiple campus locations - FR-012

2.3 User characteristics and stakeholders

Labels group stakeholders for readability. They do not combine permissions.

Education, training and experience requirements remain unspecified.

Role	Source user groups	Responsibility
Admin	network administrators, administrator	Performs only the role's source-defined operations in section 3.
Project Analyst	Document review role	Reviews requirements and tracks open decisions; this role grants no product access.

2.4 Constraints

No additional implementation constraints were confirmed. Source-defined restrictions remain in the requirements in section 3.

2.5 Assumptions and dependencies

Open decisions requiring confirmation:

- What makes a configuration change unauthorized?
- Which fields must the audit logs contain?
- What retention period is required for the audit logs?
- Which security acceptance outcomes must be demonstrated?
- What measurable reliability and recovery targets must the system meet?
- What measurable scalability targets must the system meet?
- What criteria define acceptable manageability?

3 Specific requirements

Each requirement is stated once with its original identifier and wording. Cross-references identify requirements covering more than one concern.

3.1 External interface requirements

3.1.1 User interfaces

Related requirements: FR-010. Interface details beyond those requirements remain unspecified.

3.1.2 Hardware interfaces

No specific interface requirement was supplied; confirmation is required.

3.1.3 Software interfaces

Related requirements: FR-011. Interface details beyond those requirements remain unspecified.

3.1.4 Communications interfaces

Related requirements: FR-001, FR-002, FR-003, FR-004, FR-005. Interface details beyond those requirements remain unspecified.

3.2 Functional requirements

FR-001 · Must · explicit

The system shall manage configuration changes across routers, switches, wireless controllers, and other network infrastructure.

Traceability: E-001, TEST-001

FR-002 · Should · explicit

The system shall allow Authorized network administrators to review current configurations.

Traceability: E-002, TEST-002

FR-003 · Should · explicit

The system shall allow Authorized network administrators to submit configuration changes.

Traceability: E-002, TEST-003

FR-004 · Should · explicit

The system shall allow Authorized network administrators to track change history.

Traceability: E-002, TEST-004

FR-005 · Should · explicit

The system shall allow Authorized network administrators to restore previously approved configurations when necessary.

Traceability: E-002, TEST-005

FR-006 · Should · explicit

The system shall detect unauthorized or unexpected configuration changes.

Traceability: E-003, TEST-006

FR-007 · Should · explicit

The system shall generate security alerts.

Traceability: E-003, TEST-007

FR-008 · Should · explicit

The system shall maintain audit logs of configuration activity and administrator actions.

Traceability: E-003, TEST-008

FR-009 · Should · explicit

The system shall restrict administrative functions based on approved roles.

Traceability: E-003, TEST-009

FR-010 · Should · explicit

The system shall provide a centralized dashboard.

Traceability: E-004, TEST-010

FR-011 · Should · explicit

The system shall integrate with existing university identity services.

Traceability: E-004, TEST-011

FR-012 · Should · explicit

The system shall support multiple campus locations.

Traceability: E-004, TEST-012

3.3 Performance requirements

NFR-004 · Should · explicit

The system shall remain scalable.

Traceability: E-004, TEST-016

3.4 Logical database requirements

No separate requirement was specified in the supplied brief.

3.5 Design constraints

No separate requirement was specified in the supplied brief.

3.6 Software system attributes

3.6.1 Reliability

NFR-003 · Should · explicit

The system shall remain reliable.

Traceability: E-004, TEST-015

3.6.2 Availability

No separate requirement was specified in the supplied brief.

3.6.3 Security

Also covered by: FR-007, FR-008, FR-009.

NFR-001 · Must · explicit

The system shall be secure.

Traceability: E-001, TEST-013

NFR-002 · Should · explicit

The system shall remain secure.

Traceability: E-004, TEST-014

3.6.4 Maintainability

NFR-005 · Should · explicit

The system shall remain manageable.

Traceability: E-004, TEST-017

3.6.5 Portability

No separate requirement was specified in the supplied brief.

3.7 Other requirements

No separate requirement was specified in the supplied brief.

A Derived threat hypotheses

These are defensive planning hypotheses, not observed incidents or approved extra controls. Unrecognized threats and actual implementation effectiveness require human review.

These are planning hypotheses, not observed incidents or additional approved controls.

A.1 THR-001: User identities and authenticated access

Scenario: Derived hypothesis: An attacker uses a compromised identity or authenticated session to impersonate an authorized user and exercise that user's permitted operations.

Hypothetical actor or cause: Hypothetical attacker, privileged insider or failure cause

Potential consequence: Unauthorized use of a legitimate identity could expose resources or alter system state.

Linked requirements: FR-010, FR-011, FR-012, NFR-002, NFR-003, NFR-004, NFR-005

Original evidence: E-004

Residual exposure: Unverified planning hypothesis. Linked requirements define intended behavior, not proven prevention. Validate enforcement and residual exposure during design and acceptance review; likelihood and severity are not established.

A.2 THR-002: Role and permission boundaries

Scenario: Derived hypothesis: A principal exploits an incorrect role assignment or permission check to perform operations outside the source-defined authorization boundary.

Hypothetical actor or cause: Hypothetical attacker, privileged insider or failure cause

Potential consequence: A lower-privilege user could access restricted functions or sensitive resources.

Linked requirements: FR-002, FR-003, FR-004, FR-005, FR-006, FR-007, FR-008, FR-009

Original evidence: E-002, E-003

Residual exposure: Unverified planning hypothesis. Linked requirements define intended behavior, not proven prevention. Validate enforcement and residual exposure during design and acceptance review; likelihood and severity are not established.

A.3 THR-003: Integration trust paths

Scenario: Derived hypothesis: A compromised connected system or integration path supplies forged, altered or missing identity, event or policy data used by the source-defined integration.

Hypothetical actor or cause: Hypothetical attacker, privileged insider or failure cause

Potential consequence: Untrusted integration data could cause incorrect decisions or hide security events.

Linked requirements: FR-010, FR-011, FR-012, NFR-002, NFR-003, NFR-004, NFR-005

Original evidence: E-004

Residual exposure: Unverified planning hypothesis. Linked requirements define intended behavior, not proven prevention. Validate enforcement and residual exposure during design and acceptance review; likelihood and severity are not established.

A.4 THR-004: Security and administrative audit evidence

Scenario: Derived hypothesis: An attacker or recording failure suppresses or alters source-required audit evidence so that access decisions, security events or admin actions cannot be reconstructed.

Hypothetical actor or cause: Hypothetical attacker, privileged insider or failure cause

Potential consequence: Investigators could lose reliable evidence of the events required by the brief.

Linked requirements: FR-006, FR-007, FR-008, FR-009

Original evidence: E-003

Residual exposure: Unverified planning hypothesis. Linked requirements define intended behavior, not proven prevention. Validate enforcement and residual exposure during design and acceptance review; likelihood and severity are not established.

A.5 THR-005: Monitoring and security alerts

Scenario: Derived hypothesis: Suspicious activity evades the source-defined detection workflow, or excessive events obscure actionable alerts before an authorized reviewer can investigate.

Hypothetical actor or cause: Hypothetical attacker, privileged insider or failure cause

Potential consequence: Relevant suspicious activity could remain undetected or receive delayed investigation.

Linked requirements: FR-006, FR-007, FR-008, FR-009

Original evidence: E-003

Residual exposure: Unverified planning hypothesis. Linked requirements define intended behavior, not proven prevention. Validate enforcement and residual exposure during design and acceptance review; likelihood and severity are not established.

B Risk assessment

Risk treatments are proposals for review. Unknown likelihood and impact ratings mean unassessed; no numeric confidence or severity is inferred.

B.1 RISK-001

Derived risk affecting User identities and authenticated access: Unauthorized use of a legitimate identity could expose resources or alter system state.

Likelihood: unknown; impact: unknown; owner: TBD.

Rationale: This is a planning risk linked to the defensive hypothesis: Derived hypothesis: An attacker uses a compromised identity or authenticated session to impersonate an authorized user and exercise that user's permitted operations.

Proposed treatment: Review impersonation and compromised-identity paths against the stated access duties; confirm any additional identity or session protections with the stakeholder. Assign a responsible reviewer. Verify the linked canonical obligations against their authored acceptance criteria in an authorized isolated environment. Record gaps and obtain a scope decision before adding any product behavior; preserve explicit exclusions.

Linked requirements: FR-010, FR-011, FR-012, NFR-002, NFR-003, NFR-004, NFR-005

Original evidence: E-004

B.2 RISK-002

Derived risk affecting Role and permission boundaries: A lower-privilege user could access restricted functions or sensitive resources.

Likelihood: unknown; impact: unknown; owner: TBD.

Rationale: This is a planning risk linked to the defensive hypothesis: Derived hypothesis: A principal exploits an incorrect role assignment or permission check to perform operations outside the source-defined authorization boundary.

Proposed treatment: Test permitted and denied role/resource pairs against the linked authorization duties; review ambiguous permission assignments with the resource owner. Assign a responsible reviewer. Verify the linked canonical obligations against their authored acceptance criteria in an authorized isolated environment. Record gaps and obtain a scope decision before adding any product behavior; preserve explicit exclusions.

Linked requirements: FR-002, FR-003, FR-004, FR-005, FR-006, FR-007, FR-008, FR-009

Original evidence: E-002, E-003

B.3 RISK-003

Derived risk affecting Integration trust paths: Untrusted integration data could cause incorrect decisions or hide security events.

Likelihood: unknown; impact: unknown; owner: TBD.

Rationale: This is a planning risk linked to the defensive hypothesis: Derived hypothesis: A compromised connected system or integration path supplies forged, altered or missing identity, event or policy data used by the source-defined integration.

Proposed treatment: Review trust assumptions at the specified integration boundaries with missing or altered input fixtures; confirm the required failure handling with the owner. Assign a responsible reviewer. Verify the linked canonical obligations against their authored acceptance criteria in an authorized isolated environment. Record gaps and obtain a scope decision before adding any product behavior; preserve explicit exclusions.

Linked requirements: FR-010, FR-011, FR-012, NFR-002, NFR-003, NFR-004, NFR-005

Original evidence: E-004

B.4 RISK-004

Derived risk affecting Security and administrative audit evidence: Investigators could lose reliable evidence of the events required by the brief.

Likelihood: unknown; impact: unknown; owner: TBD.

Rationale: This is a planning risk linked to the defensive hypothesis: Derived hypothesis: An attacker or recording failure suppresses or alters source-required audit evidence so that access decisions, security events or admin actions cannot be reconstructed.

Proposed treatment: Compare source-required events with the resulting audit evidence; assess missing and altered-record cases and confirm unprovided protection or retention decisions. Assign a responsible reviewer. Verify the linked canonical obligations against their authored acceptance criteria in an authorized isolated environment. Record gaps and obtain a scope decision before adding any product behavior; preserve explicit exclusions.

Linked requirements: FR-006, FR-007, FR-008, FR-009

Original evidence: E-003

B.5 RISK-005

Derived risk affecting Monitoring and security alerts: Relevant suspicious activity could remain undetected or receive delayed investigation.

Likelihood: unknown; impact: unknown; owner: TBD.

Rationale: This is a planning risk linked to the defensive hypothesis: Derived hypothesis: Suspicious activity evades the source-defined detection workflow, or excessive events obscure actionable alerts before an authorized reviewer can investigate.

Proposed treatment: Evaluate the stated detection duties with representative positive and negative events; confirm undecided thresholds, alert priorities and investigation responsibilities. Assign a responsible reviewer. Verify the linked canonical obligations against their authored acceptance criteria in an authorized isolated environment. Record gaps and obtain a scope decision before adding any product behavior; preserve explicit exclusions.

Linked requirements: FR-006, FR-007, FR-008, FR-009

Original evidence: E-003

C SDLC recommendation and delivery workflow

Proposed lifecycle: Incremental

Derived operational_validation (+6 points): Monitoring and alert behavior needs validation with operational feedback; this does not establish continuous deployment or a release frequency. (E-003) Derived deployment_scope (+12 points): Multiple deployment locations offer a staged validation and rollout opportunity; the rollout plan remains proposed. (E-004) Derived integration_assurance (+8 points): Connected systems need integration verification before broader rollout; their age, technology

and technical uncertainty remain unspecified. (E-004)

Recommendation requires human confirmation of delivery constraints and team capacity. Suitability points are deterministic rule scores, not probabilities.

Policy version: sequential-sdlc-2.0

Unconfirmed decision factors: budget, delivery, failure consequences, feedback, legacy, regulation, schedule, stability, uncertainty, verification

- Recurring delivery requires stakeholder review and backlog refinement; a cycle does not guarantee completeness or remove unresolved decisions.
- Proposed suitability recommendation; confirm delivery constraints and team capacity before adopting this lifecycle. Missing factors are not assumed.

C.1 Ranking and source evidence

- security_assurance = integrated security review (derived). Security decisions benefit from security review and tests throughout delivery. Original evidence: E-002.
- operational_validation = monitoring feedback (derived). Monitoring and alert behavior needs validation with operational feedback; this does not establish continuous deployment or a release frequency. Original evidence: E-003.
- deployment_scope = staged rollout opportunity (derived). Multiple deployment locations offer a staged validation and rollout opportunity; the rollout plan remains proposed. Original evidence: E-004.
- integration_assurance = integration validation (derived). Connected systems need integration verification before broader rollout; their age, technology and technical uncertainty remain unspecified. Original evidence: E-004.

Candidate	Suitability points
Incremental	66
Hybrid	65
DevSecOps	63
Spiral	51
Agile	40
V-Model	40
Waterfall	40

C.1.1 Incremental suitability reasons

- Derived operational_validation (+6 points): Monitoring and alert behavior needs validation with operational feedback; this does not establish continuous deployment or a release frequency. (E-003)
- Derived deployment_scope (+12 points): Multiple deployment locations offer a staged validation and rollout opportunity; the rollout plan remains proposed. (E-004)
- Derived integration_assurance (+8 points): Connected systems need integration verification before broader rollout; their age, technology and technical uncertainty remain unspecified. (E-004)

C.1.2 Hybrid suitability reasons

- Derived security_assurance (+8 points): Security decisions benefit from security review and tests throughout delivery. (E-002)
- Derived operational_validation (+5 points): Monitoring and alert behavior needs validation with operational feedback; this does not establish continuous deployment or a release frequency. (E-003)
- Derived deployment_scope (+8 points): Multiple deployment locations offer a staged validation and rollout opportunity; the rollout plan remains proposed. (E-004)
- Derived integration_assurance (+4 points): Connected systems need integration verification before broader rollout; their age, technology and technical uncertainty remain unspecified. (E-004)

C.1.3 DevSecOps suitability reasons

- Derived security_assurance (+12 points): Security decisions benefit from security review and tests throughout delivery. (E-002)
- Derived operational_validation (+8 points): Monitoring and alert behavior needs validation with operational feedback; this does not establish continuous deployment or a release frequency. (E-003)
- Derived deployment_scope (+3 points): Multiple deployment locations offer a staged validation and rollout opportunity; the rollout plan remains proposed. (E-004)

C.1.4 Spiral suitability reasons

- Derived security_assurance (+5 points): Security decisions benefit from security review and tests throughout delivery. (E-002)
- Derived integration_assurance (+6 points): Connected systems need integration verification before broader rollout; their age, technology and technical uncertainty remain unspecified. (E-004)

C.1.5 Agile suitability reasons

- No differentiating input evidence

C.1.6 V-Model suitability reasons

- No differentiating input evidence

C.1.7 Waterfall suitability reasons

- No differentiating input evidence

C.2 Methodology phases

Activities reference the canonical requirements and proposed verification work. Human gates are pending decisions, not approval records.

C.2.1 Requirements and verification planning

Proposed roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Linked requirements: FR-001, FR-002, FR-003, FR-004, FR-005, FR-006, FR-007, FR-008, FR-009, FR-010, FR-011, FR-012, NFR-001, NFR-002, NFR-003, NFR-004, NFR-005

Human gate: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

- Activity: Review and baseline the source-defined work group: FR-001: The system shall manage configuration changes across routers, switches, wireless controllers, and other network infrastructure.; FR-002: The system shall allow Authorized network administrators to review current configurations.; FR-003: The system shall allow Authorized network administrators to submit configuration changes.; FR-004: The system shall allow Authorized network administrators to track change history.; FR-005: The system shall allow Authorized network administrators to restore previously approved configurations when necessary.; FR-006: The system shall detect unauthorized or unexpected configuration changes.; FR-007: The system shall generate security alerts.; FR-008: The system shall maintain audit logs of configuration activity and administrator actions.; FR-009: The system shall restrict administrative functions based on approved roles.; FR-010: The system shall provide a centralized dashboard.; FR-011: The system shall integrate with existing university identity services.; FR-012: The system shall support multiple campus locations.
- Activity: Review and baseline the source-defined work group: NFR-001: The system shall be secure.; NFR-002: The system shall remain secure.; NFR-003: The system shall remain reliable.; NFR-004: The system shall remain scalable.; NFR-005: The system shall remain manageable.
- Activity: Create the unit, integration, system and acceptance test plan before implementation; retain each authored acceptance criterion.
- Deliverable: Versioned requirements baseline
- Deliverable: Unit, integration, system and acceptance test plan
- Deliverable: Unresolved-decision record
- Entry criterion: Authoritative inputs and canonical requirements are available for review.
- Exit criterion: A human reviewer has recorded the baseline decision and disposition of unresolved scope.
- Security check: Review implementation and acceptance evidence against FR-007: The system shall generate security alerts.
- Security check: Review implementation and acceptance evidence against NFR-001: The system shall be secure.
- Security check: Review implementation and acceptance evidence against NFR-002: The system shall remain secure.

C.2.2 Incremental delivery cycle

Proposed roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Linked requirements: FR-001, FR-002, FR-003, FR-004, FR-005, FR-006, FR-007, FR-008, FR-009, FR-010, FR-011, FR-012, NFR-001, NFR-002, NFR-003, NFR-004, NFR-005

Human gate: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

- Activity: Review and prioritize the backlog for the next iteration without changing source-defined scope.
- Activity: Implement the source-defined work group: FR-001: The system shall manage configuration changes across routers, switches, wireless controllers, and other network infrastructure.; FR-002: The system shall allow Authorized network administrators to review current configurations.; FR-003: The system shall allow Authorized network administrators to submit configuration changes.; FR-004: The system shall allow Authorized network administrators to track change history.; FR-005: The system shall allow Authorized network administrators to restore previously approved configurations when necessary.; FR-006: The system shall detect unauthorized or unexpected configuration changes.; FR-007: The system shall generate security alerts.; FR-008: The system shall maintain audit logs of configuration activity and administrator actions.; FR-009: The system shall restrict administrative functions based on approved roles.; FR-010: The system shall provide a

centralized dashboard.; FR-011: The system shall integrate with existing university identity services.; FR-012: The system shall support multiple campus locations.

- Activity: Implement the source-defined work group: NFR-001: The system shall be secure.; NFR-002: The system shall remain secure.; NFR-003: The system shall remain reliable.; NFR-004: The system shall remain scalable.; NFR-005: The system shall remain manageable.
- Activity: In an authorized isolated test environment, execute TEST-001: Given: the system manages configuration changes across routers, switches, wireless controllers, and other network infrastructure When: the system initiates a configuration change operation Expected: the resulting configuration is applied without error to the targeted device
- Activity: In an authorized isolated test environment, execute TEST-002: Given: Authorized network administrators review current configurations When: initiate the review of current configurations Expected: the requested configuration items are returned for examination
- Activity: In an authorized isolated test environment, execute TEST-003: Given: Authorized network administrators submit a configuration change When: initiate submission of a configuration change on the named object Expected: the object's resulting policy, job, submitted change, or history is observed as updated
- Activity: In an authorized isolated test environment, execute TEST-004: Given: Authorized network administrators track change history When: initiate tracking of change history on the named object Expected: the object's resulting policy, job, submitted change, or history is observed as accessible and accurate
- Activity: In an authorized isolated test environment, execute TEST-005: Given: Authorized network administrators restore previously approved configurations When: initiate restoration of a previously approved configuration on the named object Expected: the object's resulting policy, job, submitted change, or restored state is observed as valid and consistent with approval records
- Activity: In an authorized isolated test environment, execute TEST-006: Given: the system detects an unauthorized or unexpected configuration change When: the system initiates the detection process Expected: an alert is generated and logged in the audit trail
- Activity: In an authorized isolated test environment, execute TEST-007: Given: a security alert condition is triggered by an anomalous configuration event When: the system evaluates and responds to the alert condition Expected: the alert exists in the designated monitoring channel as captured externally
- Activity: In an authorized isolated test environment, execute TEST-008: Given: an administrator modifies a network configuration When: the system records the action and all associated metadata Expected: The system records the action and all associated metadata.
- Activity: In an authorized isolated test environment, execute TEST-009: Given: an administrator attempts to perform a restricted administrative function When: the system evaluates the role-based access policy Expected: the request is denied if the role lacks approved permissions
- Activity: In an authorized isolated test environment, execute TEST-010: Given: system provides a centralized dashboard When: the system initiates the display of network configuration data Expected: The system initiates the display of network configuration data and the external test capture records the resulting response.
- Activity: In an authorized isolated test environment, execute TEST-011: Given: system integrates with existing university identity services When: the system performs a login authentication request via university identity protocols Expected: an authenticated session is established and persisted across operations
- Activity: In an authorized isolated test environment, execute TEST-012: Given: system supports multiple campus locations When: the system is initiated from each stated campus location Expected: use succeeds at all locations with consistent configuration access

- Activity: In an authorized isolated test environment, execute TEST-013: Given: A draft acceptance plan for this quality goal: the system shall be secure against unauthorized access and data exfiltration via network interfaces. When: An analyst inspects or reviews that acceptance plan to determine whether measurable targets are defined or remain TBD. Expected: The plan's measurable targets remain TBD or unconfirmed, as no specific thresholds, attack scenarios, or response metrics are stated in the draft plan.
- Activity: In an authorized isolated test environment, execute TEST-014: Given: The system shall remain secure under all operational conditions. When: An analyst reviews the security assurance plan and evaluates whether it includes measurable targets for threat detection, response time, and configuration drift Expected: The plan's measurable targets remain TBD or unconfirmed
- Activity: In an authorized isolated test environment, execute TEST-015: Given: The system shall remain reliable under all operational conditions. When: An analyst reviews the reliability assurance plan and evaluates whether it includes measurable targets for failure detection, recovery time, and service uptime Expected: The plan's measurable targets remain TBD or unconfirmed
- Activity: In an authorized isolated test environment, execute TEST-016: Given: The system shall remain scalable under all operational conditions. When: An analyst reviews the scalability assurance plan and evaluates whether it includes measurable targets for throughput, latency, and resource utilization growth Expected: The plan's measurable targets remain TBD or unconfirmed
- Activity: In an authorized isolated test environment, execute TEST-017: Given: The system shall remain manageable under all operational conditions. When: An analyst reviews the manageability assurance plan and evaluates whether it includes measurable targets for configuration change frequency, audit coverage, and Expected: The plan's measurable targets remain TBD or unconfirmed
- Activity: Collect stakeholder feedback; agree its cadence with a human reviewer before scheduling cycles.
- Activity: Return accepted stakeholder feedback and observed defects to the backlog and plan the next iteration.
- Deliverable: Requirement-linked implementation increment
- Deliverable: Recorded test evidence
- Deliverable: Reviewed feedback and next-iteration backlog
- Entry criterion: Prior work, source scope and the reviewed test plan are available for iteration planning.
- Exit criterion: A human reviewer has assessed the increment, test evidence and feedback; unresolved changes return to the next iteration.
- Security check: Review implementation and acceptance evidence against FR-007: The system shall generate security alerts.
- Security check: Review implementation and acceptance evidence against NFR-001: The system shall be secure.
- Security check: Review implementation and acceptance evidence against NFR-002: The system shall remain secure.

C.2.3 Release and handoff

Proposed roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Linked requirements: FR-001, FR-002, FR-003, FR-004, FR-005, FR-006, FR-007, FR-008, FR-009, FR-010, FR-011, FR-012, NFR-001, NFR-002, NFR-003, NFR-004, NFR-005

Human gate: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

- Activity: Review the exact requirements baseline, implementation links and verification evidence before a release decision.
- Activity: Carry unresolved source decisions into handoff; generated text is not human approval.
- Deliverable: Requirements and verification evidence package
- Deliverable: Pending human release decision
- Deliverable: Outstanding-decision record
- Entry criterion: Verification evidence and unresolved decisions are available for human review.
- Exit criterion: A human has recorded the release/handoff decision for the reviewed content.
- Security check: Review implementation and acceptance evidence against FR-007: The system shall generate security alerts.
- Security check: Review implementation and acceptance evidence against NFR-001: The system shall be secure.
- Security check: Review implementation and acceptance evidence against NFR-002: The system shall remain secure.

Detailed acceptance criteria, evidence, architecture and the SDLC plan are available in the supporting report and canonical JSON.

Canonical content SHA-256: dba0d7f38f6d69894392990b489c5ee9142ebf6d916cf4a1a9fd2f425ff42969